

Variational Autoencoders

Aim:-

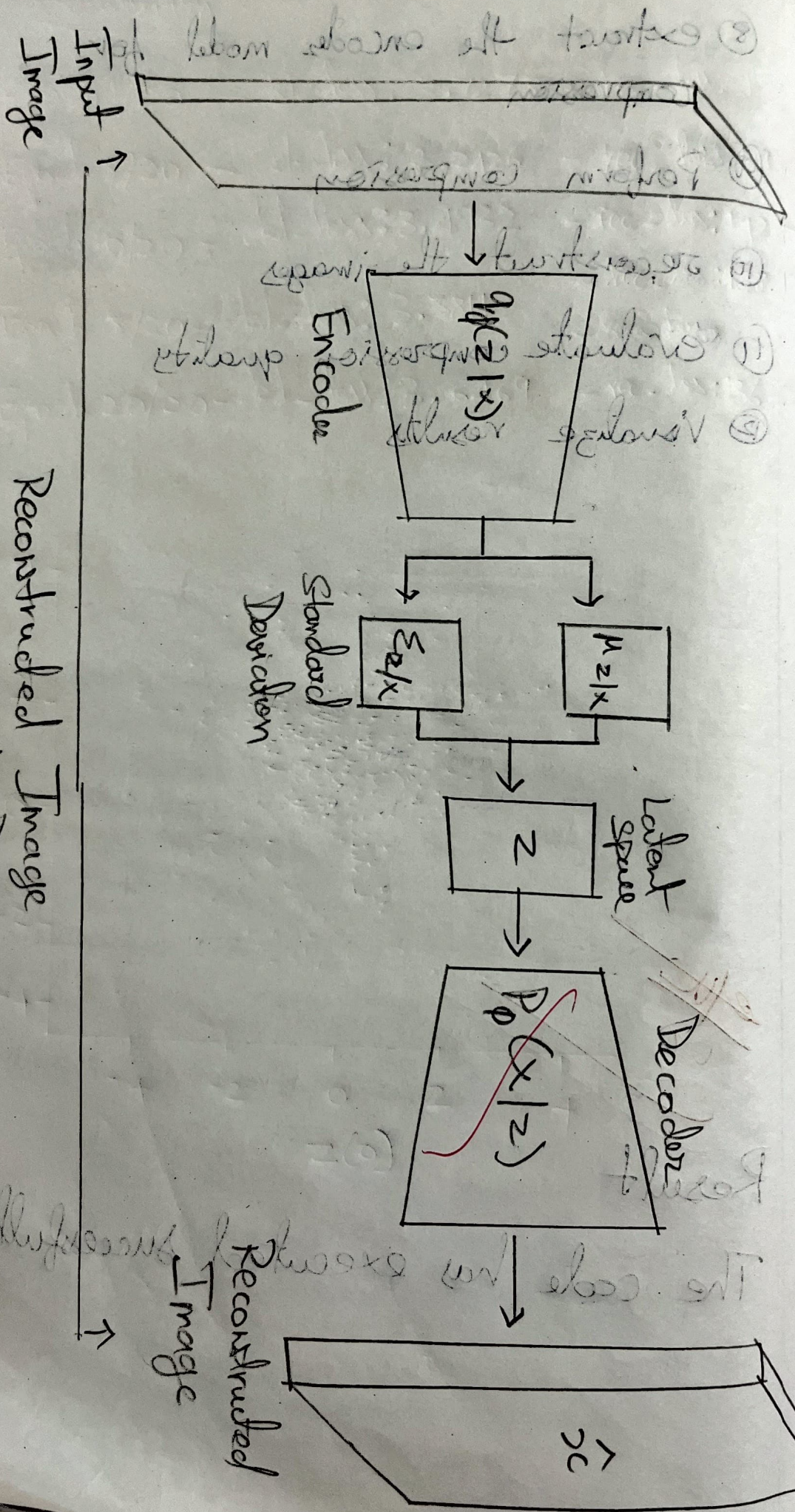
To compress and reconstruct MNIST digit images using a variational autoencoder (VAE) and to analyse its performance.

Objective:-

- ① To understand the concept of probabilistic encoding in Variational Autoencoders
- ② To implement a VAE for dimensionality reduction and image reconstruction
- ③ To train the model on MNIST and observe loss trends over epochs
- ④ To visualise reconstructed outputs and compare them with original image
- ⑤ To evaluate the stability and performance of the latent space through reconstruction error and loss graphs.

Pseudocode :-

- ① Load Mnist Dataset and normalise pixel values to $[0, 1]$



Reconstructed Image

LOSS + KL Divergence

Observation

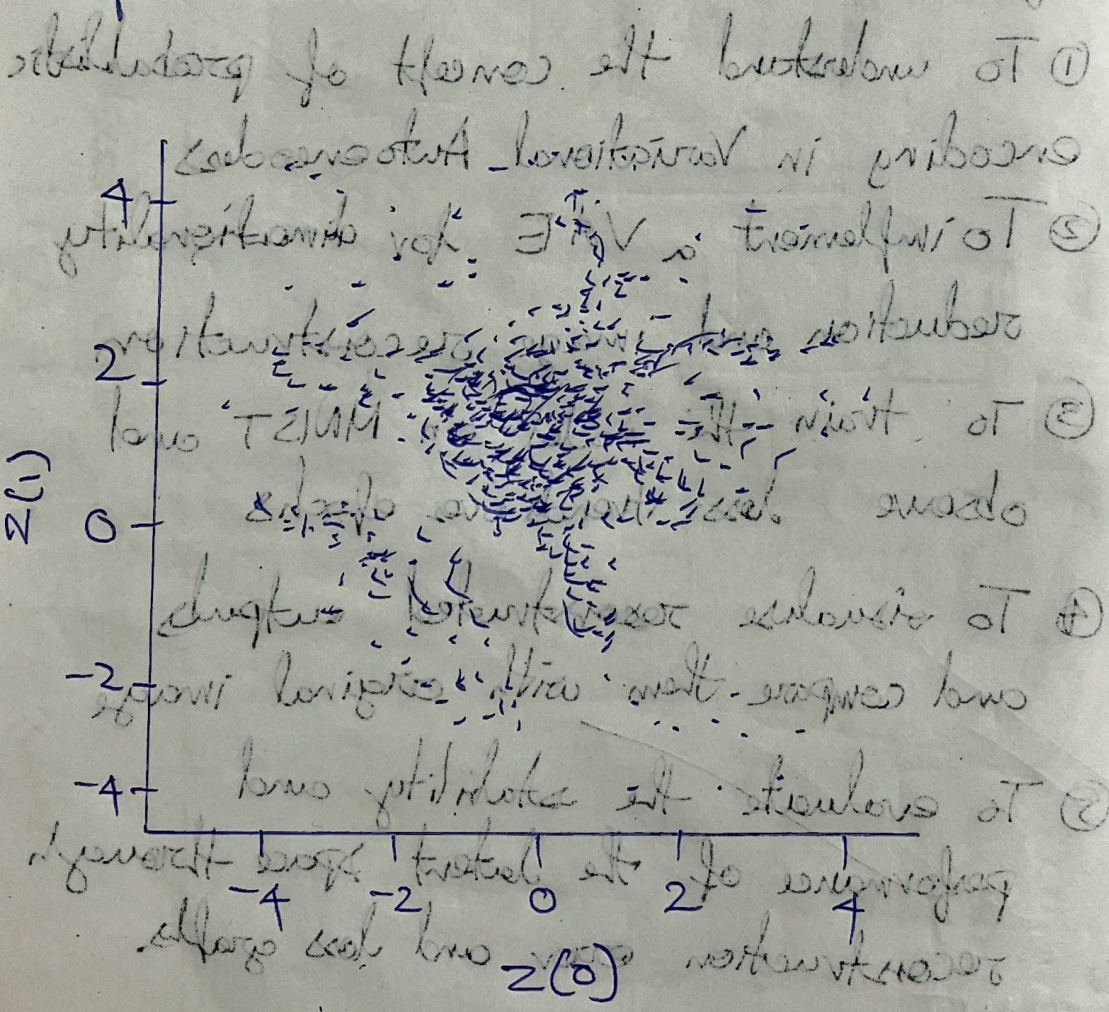
Epoch 1/20 - kl-loss: 3.7275 - rec-loss: 186.592

Epoch 5/20 - kl-loss: 3.2426 - rec-loss: 149.6942

Epoch 10/20 - kl-loss: 3.4932 - rec-loss: 142.2762

Epoch 15/20 - kl-loss: 3.5811 - rec-loss: 138.1809

Epoch 20/20 - kl-loss: 3.6469 - rec-loss: 136.2263



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2. Flatten images ($28 \times 28 \rightarrow 784$) for fully connected layers.

3. Define encoder

- Dense (512, relu) $\rightarrow$ Dense (256, relu)
- Output z -mean and z log-var (latent parameters)

4. Define decoder

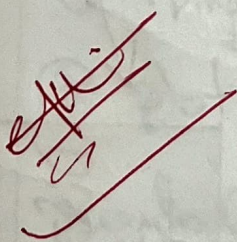
- Input z
- Dense (256, relu) $\rightarrow$ Dense (512, relu) $\rightarrow$ Dense (784, sigmoid)

5. Define VAE loss:

6. Compile and train model

7. Print epoch number and loss values

8. Plot Loss vs Epoch graph



Result

The code executed successfully